

Q. Explain types of derivation (4 Marks)

⇒ Derivation : Derivation means generating a string from the start symbol using grammar rules.

Mainly there are two types :

① Leftmost Derivation (LMD)

- leftmost non-terminal is replaced first at every step.

ex. Grammar : $S \rightarrow aSB \mid ab$
 $B \rightarrow b$

String : aabb

LMD : $S \rightarrow aSB$
 $\rightarrow aabB$
 $\rightarrow aabb$

② Rightmost Derivation (RMD)

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String : aabb

RMD : $S \rightarrow aSB$
 $\rightarrow aSb$
 $\rightarrow aabb$